Identify the problem statement

As a client, they want to predict insurance premiums. They have historical data that can be used to understand the appropriate premium based on a customer's health condition and other input data. This will help them predict future insurance charges for new customers.

Tell basic info about dataset

This is an insurance premium dataset. The input variables are **Age, Sex, BMI, Children, and Smoker**, while **Charges** is the output variable. These independent variables help predict insurance charges. The dataset contains **1,339 rows**. Since it has a relatively large number of records, the model can be trained more effectively and is likely to achieve better predictive performance.

Preprocessing Method

In this dataset, the **Sex** and **Smoker** columns contain binary categorical text values. During the data preprocessing stage, these values can be converted into numerical (ordinal) values using the **Label Encoder** algorithm. This transformation enables machine learning models to process and learn from these features effectively.

3 Stages

Stage 1: The output is expected is number so **Machine Language** algorithms will suite for this

Stage 2: **Supervised Learning**

Stage 3: **Regression**

Algorithms and Output

Multi Linear Regression

The Multi Linear Regression Use R2 Value 0.7271812748311909

Support Vector Machine

SNo	Hyper Parameter	Linear (r value)	RBF (Non Linear) (r Value)	POLY (r value)	SIGMOID
1	C10	-0.16041878058404757	-0.13961507 70443123	-0.13838527 841413395	0.139628242 78716535
2	C50	-0.1553247002693301	-0.13960686 612638007	-0.13354570 129338894	-0.13966992 240797937
3	C100	-0.15256811702292405	-0.13960288 837960477	-0.12260531 911775274	-0.13972206 98611379

The SVM Regression Use R2 Value = C10 -0.16041878058404757

Decision Tree

SNo	criterion	max_features	splitter	R_Value
1	squared_error	None	best	0.7047183281176241
2	squared_error	sqrt	best	0.6268814487283263
3	squared_error	log2	best	0.697703396856286
4	squared_error	None	random	0.7326015654271819
5	squared_error	sqrt	random	0.6702131271183249
6	squared_error	log2	random	0.5997271806402299
7	absolute_error	None	best	0.7190391039714986

8	absolute_error	sqrt	best	0.7271699534662124
9	absolute_error	log2	best	0.659826478250968
10	absolute_error	None	random	0.7383546550839297
11	absolute_error	sqrt	random	0.6394497244275421
12	absolute_error	log2	random	0.6158187977995797
13	poisson	None	best	0.696081071489306
14	poisson	sqrt	best	0.5965422514030421
15	poisson	log2	best	0.6118703928302165
16	poisson	None	random	0.6690815789054645
17	poisson	sqrt	random	0.6774328456986535
18	poisson	log2	random	0.5588880734667412

The **Decision Tree** Regression use R2 Value = **0.7383546550839297**

Random Forest

SNo	criterion	max_features	n_estimatorsint	R_Value
1	squared_error	None	10	0.7955611578286774

2	squared_error	sqrt	10	0.7987929441517292
3	squared_error	log2	10	0.7787017448836693
4	squared_error	None	50	0.8109296158832108
5	squared_error	sqrt	50	0.8171329182410264
6	squared_error	log2	50	0.8150176404284265
7	squared_error	None	100	0.8121761540654081
8	squared_error	sqrt	100	0.8187754164155994
9	squared_error	log2	100	0.81348740875662
10	absolute_error	None	10	0.8124563614123618
11	absolute_error	sqrt	10	0.8068750227248634
12	absolute_error	log2	10	0.8037267966998505
13	absolute_error	None	50	0.816141235126405
14	absolute_error	sqrt	50	0.8134177227629067
15	absolute_error	log2	50	0.7954426250704213
16	absolute_error	None	100	0.8114066316985156
17	absolute_error	sqrt	100	0.8077485164428463

18	absolute_error	log2	100	0.8070863195456356
19	poisson	None	10	0.793562068544281
20	poisson	sqrt	10	0.7979502881670042
21	poisson	log2	10	0.7879305315189464
22	poisson	None	50	0.8180697073060705
23	poisson	sqrt	50	0.8143466634991642
24	poisson	log2	50	0.8168375726114355
25	poisson	None	100	0.8139218883504525
26	poisson	sqrt	100	0.8178953383463188
27	poisson	log2	100	0.8177829771721128

The **Rain Forest** Regression use R2 Value= **0.8187754164155994**

The Final Model is **Rain Forest with** **0.8187754164155994**